

Satyam Awasthi

Ph.D. Applicant | Closed-Loop Brain-Computer Interfaces

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Summary

My research interests center on reliable closed-loop brain-computer interfaces, from physiological signal acquisition and intended-action decoding to control, feedback, and adaptation. My current work explores EEG-based interfaces and robust estimation and control under uncertainty. My graduate research at UC Santa Barbara examined eye-tracking behavior and calibration during mixed-reality locomotion.

Education

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| <p>M.S. University of California, Santa Barbara, Computer Science</p> <ul style="list-style-type: none"> GPA: 3.91/4.00 Graduate research in the Four Eyes Laboratory with Prof. Tobias Höllerer and Prof. Michael Beyeler. Studied gaze-tracking reliability and calibration during mixed-reality locomotion through EyeTTS. | <p>Santa Barbara, CA
Sept 2021 – June 2023</p> |
| <p>B.Tech. Indian Institute of Technology, Kharagpur, Electrical Engineering</p> <ul style="list-style-type: none"> GPA: 8.98/10.0; Minor in Computer Science. Core technical member of the Autonomous Ground Vehicle and Swarm Robotics research groups. Projects in wearable sensing, human activity recognition, autonomous navigation, and swarm coordination. | <p>Kharagpur, India
July 2016 – July 2020</p> |

Interests

Closed-loop brain-computer interfaces: Physiological signal acquisition, Intended-action decoding, Control, feedback, and adaptation

Reliable inference and control: Robust output-feedback MPC, State estimation under imperfect sensing, Confidence-aware decisions

Human sensing and interaction: EEG and movement kinematics, Eye-tracking reliability and calibration, Inertial sensing and movement reconstruction

Research Questions

- How much can noisy physiological signals tell us about an intended action, and how early and precisely can we infer it?
- How should a system act when that inference is uncertain?

Research Experience

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| <p>Inria, SSVEP-Based EEG Navigation — ongoing research under guidance</p> <p>Under the guidance of Prof. Camille Gontier.</p> <ul style="list-style-type: none"> Developing a four-channel EEG interface for confidence-aware navigation in a C++/OpenGL maze. Assembled BioAmp acquisition hardware with shared ground and reference; implemented visual targets and raw acquisition diagnostics. Built photodiode capture hardware and Python tools to measure delivered stimulus timing. Calibrated EEG streaming, CCA decoding, and EEG-controlled navigation remain integration and evaluation work. | <p>2026 – present</p> |
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University of Reading, EEG-to-Kinematics Decoding — ongoing research under guidance Under the guidance of Prof. Slawomir Nasuto. • Predicting wrist X, Y, and Z position from 32-channel EEG using the WAY-EEG-GAL dataset. • Built run-separated evaluation with training-only scaling and windows that never cross recording-run boundaries. • Compared six regression baselines on held-out runs from participant P1; CNN-BiLSTM achieved R^2 of 0.809, 0.783, and 0.748. • Results cover one participant and one seed; cross-participant transfer and broader validation remain future work.	2026 – present
Autonomous Systems Group, Imperial College London , Research Collaboration — State-Dependent Robust Output-Feedback MPC Under the guidance of Prof. Johannes Köhler. • Developing dynamic estimation and tracking-error bounds for state-dependent uncertainty using a reproduced ROHMPc certificate. • Reused the offline design’s metric, observer gain, feedback gain, and LMI certificate to propagate coupled error bounds. • Illustrated a 6.68% final-time observer-radius reduction under a prescribed synthetic disturbance envelope. • The result is a tube-propagation illustration; closed-loop MPC integration and validation remain ongoing.	July 2026 – present
Four Eyes Laboratory, University of California, Santa Barbara , Graduate Researcher — EyeTTS Advisor: Prof. Tobias Höllerer; co-advisor: Prof. Michael Beyeler. • Developed a Unity-based eye-tracking study platform and Python calibration and analysis pipeline for mixed-reality locomotion. • Studied gaze errors across target motion, head/body movement, and spatial reference frames. • Contributed the Magic Leap 1 study framework and calibration work; Vivian Ross and Sydney Lim developed the HoloLens 2 and Meta Quest Pro frameworks. • First-author poster papers appeared at IEEE ISMAR Adjunct 2023 and IEEE VR Workshops 2024.	Sept 2021 – June 2023
Autonomous Ground Vehicle & Swarm Robotics Group, IIT Kharagpur , Undergraduate Researcher • Contributed perception, multimodal state-estimation, and navigation components to Eklavya, the team’s autonomous ground vehicle. • Worked on visual perception, hierarchical motion planning, local collision avoidance, and EKF-based sensor fusion in ROS. • The Eklavya team finished runner-up at IGVC 2018.	Jan 2016 – Dec 2018

Publications

Eye Tracking Performance in Mobile Mixed Reality First-author poster paper, pp. 1049–1050. Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, Tobias Höllerer 10.1109/VRW62533.2024.00321 (2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW))	2024
EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion First-author poster paper, pp. 479–480. Satyam Awasthi, Vivian Ross, Michael Beyeler, Tobias Höllerer 10.1109/ISMAR-Adjunct60411.2023.00104 (2023 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct))	2023

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing

2022

Collaborative work on handwriting reconstruction from smartwatch inertial signals, pp. 675–683.

Snigdha Das, Satyam Awasthi, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, Bivas Mitra

[10.1109/COMSNETS53615.2022.9668531](https://doi.org/10.1109/COMSNETS53615.2022.9668531) (2022 14th International Conference on COMMunication Systems & NETworkS (COMSNETS))

Industry Experience

Intuit Inc, Software Engineer 2

Mountain View, CA
Mar 2024 – Jan 2026

- Built dual-stage agent pipelines for intent routing, structured tool execution, and generative user-interface composition.
- Developed production platform infrastructure supporting large-scale client applications and internal developer workflows.

Yahoo Inc, Software Dev Engineer II (IC3)

Mountain View, CA
Apr 2023 – Nov 2023

- Developed production Android systems for Yahoo Mail, including data-driven inbox commerce and expense-management features.

Coupang Global LLC, Software Engineering Intern

Mountain View, CA
June 2022 – Sept 2022

- Developed user-facing search and product-discovery features for Coupang's Android commerce platform.

Samsung Research Institute, Software Engineer (CL 2)

Noida, India
Sept 2020 – Sept 2021

- Developed Android platform and device applications involving media processing, privacy, and isolated user-storage architectures.

Teaching

University of California, Santa Barbara, Graduate Teaching Assistant

Santa Barbara, CA
Sept 2021 – June 2023

Six quarters of teaching across introductory, mobile-development, and capstone courses; mentored and evaluated 300+ students.

- Led weekly discussion sections for up to 60 students; designed coding assignments, lab tests, quizzes, and grading criteria.
- Delivered guest lectures and guided senior capstone projects from ideation to deployment.
- CMPSC 16: Problem Solving with Computers I — Fall 2021 and Winter 2022.
- CMPSC 184: Mobile Application Development — Spring 2022.
- CMPSC 24: Problem Solving with Computers II — Summer 2022.
- CMPSC 189A/B: Senior Computer Systems Project — Fall 2022 and Winter 2023.

Projects

SmartWatch for Wall Writing (WatchScribe)

2019 – 2020

IIT Kharagpur. Reconstructing wall writing from smartwatch accelerometer and gyroscope signals.

- Combined rotational kinematics, orientation correction, and unsupervised stroke separation without a pretrained handwriting model.
- Evaluated reconstructed writing through human readability and geometric comparison; published at COMSNETS 2022.

JitterNot

Spring 2022

UC Santa Barbara. Learning-assisted model predictive control for adaptive video streaming.

- Trained an LSTM throughput predictor offline and used its online forecasts to select video-resolution actions.
- Evaluated constrained QoE optimization against Jitsi Meet's default adaptation under emulated network conditions.

Wearable Human Activity Recognition

Nov 2019

IIT Kharagpur. DeepConvLSTM modeling of smartphone inertial signals for human-motion recognition.

- Streamed six sensor channels at 100 Hz to a server for inference on overlapping five-second windows.
- Combined TimeDistributed convolutional feature extraction with LSTM sequence modeling.

Dual-Stage Agent Pipelines for Generative UI

Winter 2025

Intuit. Separating user-intent interpretation and state tracking from structured UI generation.

- Used structured outputs, Pydantic/JSON Schema validation, and retrieved component primitives to constrain generated layouts.

CohesiveAR

Fall 2022

UC Santa Barbara. Texture extraction and homography mapping for augmented reality.

- Combined ARCore surface localization, OpenCV perspective rectification, and dynamic texture mapping onto virtual surfaces.

Spell Training VR

Winter 2022

UC Santa Barbara. Mid-air gesture interaction for a Unity-based VR game.

- Connected webcam-based hand tracking and sketch recognition to Unity using WebSockets.

Battleship Solitaire Solver

Spring 2022

UC Santa Barbara. Solver-aided constraint synthesis in Rosette/Racket.

- Encoded fleet composition, initial clues, non-adjacency, and row/column tallies as symbolic constraints.
- Built a puzzle generator and evaluated solving time across grid sizes, fleet configurations, and packing fractions.

Skills

Programming and scientific computing: Python, C++, MATLAB, Java, Kotlin, Rosette/Racket

Machine learning and signal processing: PyTorch, NumPy/SciPy, OpenCV, CNN/LSTM models, Time-series modeling, Spectral analysis, Digital filtering, Wearable/inertial signal processing, Run-separated evaluation

Control and estimation: Model predictive control, Robust output-feedback MPC, Contraction-based methods, State estimation (EKF), Sensor fusion, Motion planning

Robotics and interactive systems: ROS/ROS2, Gazebo, RViz, Unity, OpenGL, ARCore, BioAmp acquisition, Optical stimulus measurement

Developer tools: Git

Academic Service

- Peer Reviewer, ACM UIST 2024 — reviewed submissions on interactive systems, spatial tracking, and interface hardware.